Towards One-Step Flow Matching: A Survey

Nhat H. Tran
Hanoi University of Science and Technology
Hanoi, Vietnam
nhat.th261124m@sis.hust.edu.vn

Luong T. Nguyen
Hanoi University of Science and Technology
Hanoi, Vietnam
luong.nt252743m@sis.hust.edu.vn

Alisa S. Hamidova
Hanoi University of Science and Technology
Hanoi, Vietnam
alisa.hs252893m@sis.hust.edu.vn

Abstract

Flow matching has emerged as a powerful simulation-free framework for continuous generative modeling, yet producing a single high-quality sample still requires tens to hundreds of neural function evaluations. This survey traces the rapidly evolving push toward one-step flow matching: the distillation of an entire generative trajectory into a single network call. We begin by reviewing the foundations of flow matching and conditional flow matching, then examine traditional acceleration strategies including cache-based sampling and step distillation. We next discuss closely related paradigms such as consistency models and flow-map distillation, which provide the conceptual stepping stones for direct noise-to-data transport. The core of the survey is devoted to three modern one-step families—consistency flow matching, mean flows, and terminal velocity matching—analyzing their training objectives, theoretical guarantees, and practical trade-offs. We further summarize experimental setups, standard baselines, and state-of-the-art results, showing that the quality gap between one-step and many-step generation has narrowed substantially. By unifying these developments under a single narrative, we highlight the field's progression from simulation-based sampling to direct prediction of the transport map, and provide practical guidance for researchers entering efficient generative modeling.

1 Introduction

Generative modeling has undergone a profound transformation with the advent of diffusion models and their continuous-time counterparts, flow-based models [Ho et al., 2020, Song et al., 2021, Lipman et al., 2023]. These approaches construct a transport between a simple noise distribution and a complex data distribution by learning a time-dependent velocity field that governs the evolution of probability mass through an ordinary differential equation (ODE). While diffusion models have achieved state-of-the-art results in image synthesis, audio generation, and scientific simulation, they share a critical limitation: generating a single sample requires simulating the ODE through many time steps, each demanding a forward pass of a large neural network. For state-of-the-art Flow Matching models, this can translate to 50–1000 function evaluations (NFEs), rendering real-time or interactive deployment impractical.

Flow Matching (FM) and its most common variant, Conditional Flow Matching (CFM) [Lipman et al., 2023], provide an especially elegant formulation of this family. By regressing vector fields of fixed conditional probability paths, FM offers a simulation-free training objective for continuous normalizing flows and yields straight, optimal-transport trajectories that are faster to integrate than the curved paths typical of diffusion models. Beyond images, FM and CFM have found broad application

Figure 1: One-step samples from Terminal Velocity Matching on both ImageNet-512×512 and ImageNet-256×256 [Zhou et al., 2026].

in speech synthesis, where they power neural vocoders and text-to-speech systems by mapping latent noise directly to raw waveforms or mel-spectrogram features; they are also used in music generation, video synthesis, protein design, and molecule generation. These successes make flow matching a natural and timely lens through which to study efficient generative modeling.

This computational bottleneck has catalyzed extensive research toward one-step and few-step generative models. The central question is whether it is possible to distill the expensive many-step sampling process into a direct map that transports noise to data in a single neural network evaluation, while preserving the quality, diversity, and controllability of the original model. Early approaches to this problem, such as caching intermediate computations, progressive distillation [Salimans and Ho, 2022], and consistency models [Song et al., 2023], provided proof-of-concept results but struggled with scalability, multi-step degradation, or the need for pre-trained teachers.

More recently, flow matching has provided a unified lens for understanding these acceleration efforts. It frames generative modeling as learning a probability flow ODE, and one-step methods can be seen as different strategies for bypassing the costly simulation of that ODE—whether by enforcing consistency along trajectories, predicting average velocities, or matching terminal dynamics. Despite their differing objectives, these approaches share a common aim: replacing iterative sampling with direct, single-step or few-steps prediction.

This survey traces the development of one-step flow matching from its conceptual origins to the current state of the art. Our contributions are as follows:

- We provide a unified narrative that connects flow matching, traditional acceleration techniques, consistency models, and modern distillation methods within a single mathematical framework.
- We analyze three key methodological advances—consistency flow matching, mean flows, and terminal velocity matching—comparing their training requirements, theoretical guarantees, and empirical performance.
- We summarize experimental setups and standard baselines used to evaluate one-step models, placing state-of-the-art results in a unified comparative context.
- We highlight practical trade-offs between training from scratch and distilling from a teacher, and identify open challenges for real-world deployment.

The remainder of this survey is organized as follows. Section 2 reviews related work, including the foundations of flow matching and conditional flow matching (section 2.1), traditional acceleration strategies such as caching and step distillation (section 2.2), and related paradigms including consistency models (section 2.3). Section 3 presents the three core families of one-step methods: consistency flow matching, mean flows, and terminal velocity matching. Section 4 describes the experimental setup, datasets, metrics, and baselines. Section 5 synthesizes the empirical results. Section 7 concludes with future directions.

Algorithm 1 Flow Matching Training

Require: Data x_1, noise x_0, velocity network v_θ, optimizer
Sample $t \sim \text{Uniform}(0, 1)$
$x_t \leftarrow (1 - t)\, x_0 + t\, x_1$
$u_t \leftarrow x_1 - x_0$ ▷ Conditional velocity for OT path
$v_{\text{pred}} \leftarrow v_\theta(x_t, t)$
$\mathcal{L} \leftarrow \|v_{\text{pred}} - u_t\|^2$
Update θ with $\nabla_\theta \mathcal{L}$
return $\mathcal{L}$

2 Related Work

Flow matching provides the mathematical foundation for the one-step methods surveyed in this paper. In this section, we review the core ideas of flow matching and conditional flow matching, trace traditional strategies for accelerating diffusion and flow models, and situate the modern one-step literature among closely related paradigms.

2.1 Flow Matching and Conditional Flow Matching

Flow Matching (FM) [Lipman et al., 2023] introduces a simulation-free paradigm for training Continuous Normalizing Flows (CNFs). The key idea is to learn a time-dependent velocity field $v_\theta(t, x)$ whose associated probability flow ODE transports a simple source distribution p_0 (typically Gaussian noise) to a target data distribution p_1:

$$\frac{d}{dt} x_t = v_\theta(t, x_t), \qquad x_0 \sim p_0, \quad x_1 \sim p_1. \tag{1}$$

Rather than simulating this ODE during training, FM defines a conditional path $x_t = \alpha_t x_0 + \beta_t x_1$ between each data-noise pair (x_0, x_1) and regresses the network prediction against the corresponding conditional velocity $u_t = \dot{\alpha}_t x_0 + \dot{\beta}_t x_1$.

The marginal velocity field is the expectation of u_t over all (x_0, x_1) pairs consistent with x_t, which is intractable to compute directly. Conditional Flow Matching (CFM) sidesteps this difficulty by using the conditional objective

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t,x_0,x_1} \|v_\theta(x_t, t) - u_t\|^2, \tag{2}$$

which shares the same gradient with respect to θ as the intractable marginal objective while remaining computationally efficient [Lipman et al., 2023]. The minimizer of eq. (2) equals the true marginal velocity field, so training amounts to standard regression on sampled (t, x_0, x_1) tuples.

A particularly important choice is the *optimal transport (OT) path*, where $\alpha_t = 1 - t$ and $\beta_t = t$. This yields the straight-line interpolant $x_t = (1 - t)x_0 + tx_1$ and a constant conditional velocity $u_t = x_1 - x_0$. These straight paths minimize transport cost, enable faster training, and have been shown to improve generalization and sample quality compared with the curved trajectories typical of diffusion models [Lipman et al., 2023].

Flow matching thus provides the training backbone for all subsequent methods in this survey: it defines the probability flow ODE that one-step methods seek to compress, and the conditional regression objective that underlies both standard and one-step training.

2.2 Traditional Methods

Before the emergence of dedicated one-step flow matching, practitioners accelerated generative models through two broad strategies: reducing the cost of each function evaluation via caching, and reducing the number of evaluations via step distillation. Both approaches predate modern one-step FM but remain relevant as baselines and complementary techniques.

2.2.1 Cache-based acceleration

Because Conditional Flow Matching is typically sampled with many-step ODE solvers, its inference pipeline resembles that of a diffusion model: a deep network is evaluated sequentially, and adjacent

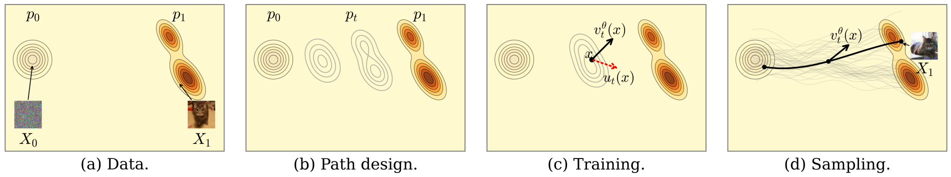

Figure 2: The Flow Matching blueprint. (a) The goal is to find a flow mapping samples X_0 from a known source or noise distribution p into samples X_1 from an unknown target or data distribution q. (b) To do so, design a time-continuous probability path $(p_t)_{0\leq t\leq 1}$ interpolating between $p := p_0$ and $q := p_1$. (c) During training, use regression to estimate the velocity field u_t known to generate p_t. (d) To draw a novel target sample $X_1 \sim q$, integrate the estimated velocity field $u_t^\theta(X_t)$ from $t = 0$ to $t = 1$, where $X_0 \sim p$ is a novel source sample.

evaluations often produce highly similar intermediate activations. Consequently, flow matching can directly adopt training-free caching methods originally developed for diffusion.

Formally, one Euler step of a flow-matching sampler can be written as

$$x_{t+\Delta t} = x_t + \Delta t\, v_\theta(x_t, t), \tag{3}$$

where $v_\theta(x_t, t)$ is the neural velocity estimate. If we decompose the network into a feature extractor ϕ_θ and a read-out head g_θ, so that $v_\theta(x_t, t) = g_\theta\big(\phi_\theta(x_t, t)\big)$, then caching stores the features $h_t = \phi_\theta(x_t, t)$ from a previous step and reuses them when the timestep changes slowly:

$$\tilde{v}_\theta(x_t, t) = g_\theta\big(h_{t-\delta}\big). \tag{4}$$

DeepCache [Ma et al., 2023] is the canonical example: it caches high-level U-Net features across denoising steps and updates only low-level features, yielding large speedups on Stable Diffusion and latent diffusion models with minimal quality loss. More recent diffusion caching schemes such as FasterCache [Lv et al., 2024] and TeaCache [Liu et al., 2025] refine this idea for video models by exploiting redundancy in conditional/unconditional branches and timestep-aware feature reuse, respectively. Because these methods operate on the backbone architecture rather than the sampling distribution, they can be ported to flow-matching image and video generators with little or no modification.

Work that explicitly targets flow matching caching is still scarce. MeanCache [Gao et al., 2026] is a recent exception: it argues that instantaneous-velocity feature caching introduces trajectory drift, and instead caches Jacobian–vector products to construct interval average velocities for more stable FM acceleration. Outside of vision, predictive feature caching has been applied to flow-matching models of molecular geometry generation [Sommer et al., 2025], and layer-selective attention caching has been used to accelerate flow-matching transformers for audio separation. These domain-specific works confirm that caching remains a valuable, largely model-agnostic acceleration tactic. Nevertheless, caching lowers the cost per step rather than the step count, so it is complementary to—rather than a replacement for—the one-step methods surveyed below.

2.2.2 Step distillation

Step distillation trains a student model that requires fewer sampling steps than its teacher. The prototypical approach is progressive distillation [Salimans and Ho, 2022], which iteratively halves the number of sampling steps by distilling a $2N$-step teacher into an N-step student. Given a teacher sampler $f_{\theta^*}(\cdot, \cdot; 2N)$ that produces an estimate after $2N$ function evaluations, progressive distillation minimizes

$$\mathcal{L}_{\text{PD}}(\theta) = \mathbb{E}_{t,x_t}\Big[\big\|f_\theta(x_t, t; N) - f_{\theta^*}(x_t, t; 2N)\big\|^2\Big], \tag{5}$$

where $f_\theta(x_t, t; N)$ denotes the student output obtained with N steps. Although effective for moderate speedups, progressive distillation is expensive—the distillation procedure must be repeated for each halving—and can suffer from error accumulation as the step budget shrinks.

A closely related idea is rectified flow [Liu et al., 2022], which learns straight ODE trajectories by refitting the transport map between noise and data. It can be viewed as minimizing the same conditional objective as FM but with the goal of straightening the coupling:

$$\mathcal{L}_{\mathrm{RF}}(\theta) = \mathbb{E}_{t,x_0,x_1}\left[\left\|v_\theta\big((1-t)x_0 + tx_1, t\big) - (x_1 - x_0)\right\|^2\right]. \tag{6}$$

Repeated rectification yields increasingly straight flows, and when the flow becomes exactly straight a single Euler step gives exact transport. InstaFlow [Liu et al., 2024] combined rectified flow with distillation to produce the first Stable-Diffusion-quality one-step text-to-image model, demonstrating that flow-based distillation could outperform progressive distillation on standard benchmarks.

These methods established the feasibility of few-step and one-step sampling, but they also revealed key limitations—expensive multi-stage training, dependence on high-quality teachers, and difficulty in scaling to high resolutions—that motivated the dedicated one-step flow matching methods discussed in section 3.

2.3 Related Methods

Two closely related lines of work set the stage for modern one-step flow matching: consistency models, which enforce that all points on a trajectory map to the same endpoint, and flow-map distillation, which learns the general transport operator between arbitrary times.

2.3.1 Consistency Models

Consistency Models (CMs) [Song et al., 2023] train a network $f_\theta(x_t, t)$ that maps any point on a diffusion or flow trajectory directly to the trajectory's endpoint x_0. The defining consistency property requires that $f_\theta(x_t, t) = f_\theta(x_{t'}, t')$ whenever x_t and $x_{t'}$ lie on the same trajectory. CMs demonstrated that one-step generation was possible without iterative sampling, but early variants struggled with scalability, multi-step error accumulation, and the need for careful curriculum training. These limitations motivated the consistency-flow-matching approaches discussed in section 3.

2.3.2 Flow-map distillation (out of scope)

A related but distinct direction is flow-map distillation, which generalizes the consistency-model idea from endpoint prediction to learning the *flow map* $f_\theta(x_t, t, s)$ that transports a state from time t to any other time s along the teacher ODE. Methods such as Align Your Flow [Sabour et al., 2025] have scaled this idea to high-resolution image generation. Because flow-map methods operate on a different design axis than the velocity-based methods that are the focus of this survey, we do not discuss them in detail; we note them here for completeness and refer interested readers to the original works.

3 One-Step Generative Modeling

Building on the flow matching framework, recent methods have developed increasingly sophisticated techniques for learning one-step or few-step generators. We discuss three representative velocity-based approaches that span the spectrum from training from scratch to distillation from pre-trained models. Flow-map distillation methods are related but outside the scope of this survey; we noted them briefly in section 2.3.

3.1 Consistency Flow Matching

Consistency Flow Matching (Consistency-FM) [Zhou et al., 2024] proposes to enforce *velocity consistency* along flow trajectories, directly defining straight flows without requiring optimal transport coupling or expensive transport plans.

Given a flow trajectory $\gamma_x(t)$ with velocity $v(t, \gamma_x(t))$, the consistency condition requires the velocity to be constant along the trajectory:

$$v(t, \gamma_x(t)) = v(s, \gamma_x(s)) \quad \forall t, s \in [0, 1]. \tag{7}$$

This is equivalent to the trajectory condition $\gamma_x(t) + (1-t)v(t, \gamma_x(t)) = \gamma_x(s) + (1-s)v(s, \gamma_x(s))$, which implies that the quantity $f(t, x_t) = x_t + (1-t)v_\theta(t, x_t)$ is constant along the path. The

Algorithm 2 Consistency Flow Matching Training

Require: Data x_1, noise x_0, velocity network v_θ, EMA model v_{θ^-}, optimizer, hyperparameter α

Sample $t \sim \text{Uniform}(0, 1 - \Delta t)$
$x_t \leftarrow (1-t)\, x_0 + t\, x_1$
$x_{t+\Delta t} \leftarrow (1 - (t + \Delta t))\, x_0 + (t + \Delta t)\, x_1$
$v_t \leftarrow v_\theta(x_t, t)$
$v_{t+\Delta t} \leftarrow v_\theta(x_{t+\Delta t}, t + \Delta t)$
$f_t \leftarrow x_t + (1-t)\, v_t$
$f_{t+\Delta t} \leftarrow x_{t+\Delta t} + (1 - (t + \Delta t))\, v_{t+\Delta t}$
▷ with no grad
$f^{\text{ema}}_{t+\Delta t} \leftarrow x_{t+\Delta t} + (1 - (t + \Delta t))\, v_{\theta^-}(x_{t+\Delta t}, t + \Delta t)$
$f^{\text{ema}}_t \leftarrow x_t + (1-t)\, v_{\theta^-}(x_t, t)$
$\mathcal{L}_f \leftarrow \|f_t - f^{\text{ema}}_{t+\Delta t}\|^2 + \|f_{t+\Delta t} - f^{\text{ema}}_t\|^2$
$\mathcal{L}_v \leftarrow \|v_t - v_{\theta^-}(x_{t+\Delta t}, t + \Delta t)\|^2 + \|v_{t+\Delta t} - v_{\theta^-}(x_t, t)\|^2$
$\mathcal{L} \leftarrow \mathcal{L}_f + \alpha\, \mathcal{L}_v$
Update θ with $\nabla_\theta \mathcal{L}$; update EMA model
return $\mathcal{L}$

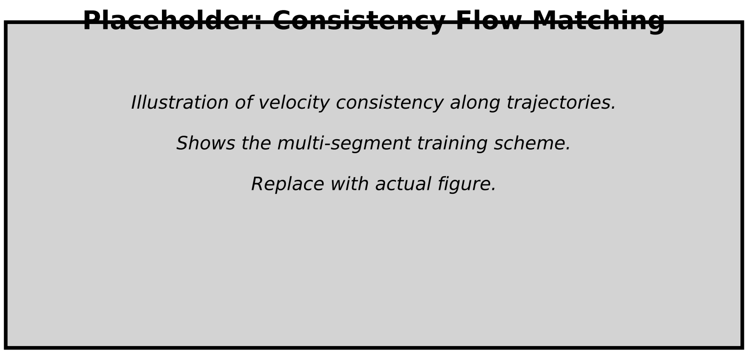

Figure 3: Placeholder: Illustration of Consistency Flow Matching velocity consistency. Replace with a figure showing the multi-segment training scheme.

training objective is:

$$\mathcal{L}_\theta = \mathbb{E}_{t, x_t, x_{t+\Delta t}} \Big[\|f_\theta(t, x_t) - f_{\theta^-}(t + \Delta t, x_{t+\Delta t})\|^2 + \alpha \|v_\theta(t, x_t) - v_{\theta^-}(t + \Delta t, x_{t+\Delta t})\|^2 \Big], \quad (8)$$

where θ^- denotes an exponentially moving average of the parameters. The first term enforces consistency in the endpoint prediction, while the second term regularizes the velocity field.

For complex distributions where a single straight path is insufficient, Consistency-FM introduces *multi-segment training*, which divides the time interval $[0, 1]$ into K segments and learns a consistent velocity v^i_θ within each segment. This allows piece-wise linear trajectories that can better approximate curved transport paths.

3.2 Mean Flows

Mean Flow [Geng et al., 2025] addresses a fundamental limitation of previous methods: the reliance on pre-trained teacher models or expensive distillation pipelines. It proposes a principled framework for one-step generative modeling that can be trained from scratch.

The key insight is to model the *average velocity* rather than the instantaneous velocity. For a trajectory z_t, the average velocity between times r and t is defined as:

$$u(z_t, r, t) = \frac{1}{t - r} \int_r^t v(z_\tau, \tau)\, d\tau. \quad (9)$$

Algorithm 3 MeanFlow Training

Require: Data x, noise ϵ, average-velocity network u_θ, optimizer
Sample $r, t \sim \text{Uniform}(0, 1)$ with $r < t$
$z_t \leftarrow (1-t)\,x + t\,\epsilon$ ▷ Interpolant
$v \leftarrow \epsilon - x$ ▷ Instantaneous conditional velocity
$u_{\text{pred}} \leftarrow u_\theta(z_t, r, t)$
Compute JVP: $(v, 1) \cdot \nabla u_\theta(z_t, r, t)$ ▷ Total derivative $\frac{d}{dt}u$
$u_{\text{tgt}} \leftarrow v - (t-r) \cdot \frac{d}{dt}u$ ▷ MeanFlow Identity
$\mathcal{L} \leftarrow \|u_{\text{pred}} - \text{sg}(u_{\text{tgt}})\|^2$
Update θ with $\nabla_\theta \mathcal{L}$
return $\mathcal{L}$

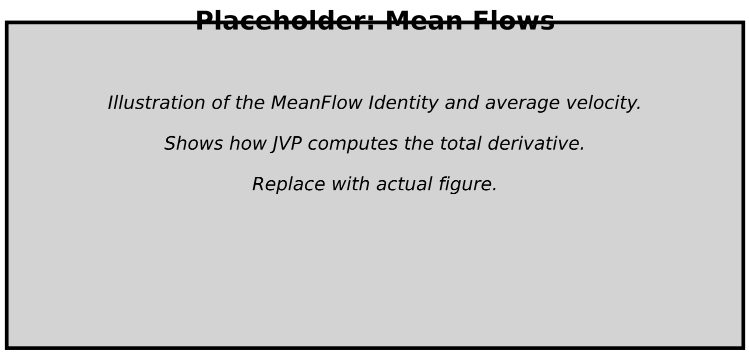

Figure 4: Placeholder: Illustration of the MeanFlow Identity and average velocity. Replace with a figure showing how JVP computes the total derivative.

Differentiating the definition $(t-r)u = \int_r^t v\, d\tau$ with respect to t yields the *MeanFlow Identity*:

$$u(z_t, r, t) = v(z_t, t) - (t-r)\frac{d}{dt}u(z_t, r, t), \tag{10}$$

where the total derivative $\frac{d}{dt}u = v \cdot \partial_z u + \partial_t u$ is computed efficiently as a Jacobian-Vector Product (JVP).

The training objective regresses the network prediction u_θ against the target derived from the MeanFlow Identity:

$$\mathcal{L}(\theta) = \mathbb{E}\|u_\theta(z_t, r, t) - \text{sg}(v(z_t, t) - (t-r)(v(z_t, t) \cdot \partial_z u_\theta + \partial_t u_\theta))\|^2, \tag{11}$$

where sg denotes the stop-gradient operator. Importantly, MeanFlow trains entirely from scratch without distillation or curriculum learning, and achieves state-of-the-art results: FID 3.43 with a single function evaluation on ImageNet 256×256.

3.3 Terminal Velocity Matching

Terminal Velocity Matching (TVM) [Zhou et al., 2026] provides a generalization of flow matching that directly learns the ODE integral while offering strong theoretical guarantees. Instead of matching time derivatives at the initial time, TVM matches them at the *terminal time* of trajectories.

Define the displacement map $f(x_t, t, s) = \int_t^s u(x_r, r)\, dr$. The terminal velocity condition states:

$$\frac{d}{ds}f(x_t, t, s) = u(\psi(x_t, t, s), s), \tag{12}$$

where ψ denotes the ODE solution map. The key theorem shows that the terminal velocity error upper-bounds the displacement error:

$$\mathcal{L}^t_{\text{displ}}(\theta) \leq \int_0^t \mathbb{E}_{x_t}\left[\|\frac{d}{ds}f_\theta(x_t, t, s) - u(\psi(x_t, t, s), s)\|^2\right] ds. \tag{13}$$

Algorithm 4 Terminal Velocity Matching (TVM) Training

Require: Data x, noise ϵ, displacement network F_θ, velocity network u_θ, optimizer

Sample $t, s \sim \text{Uniform}(0, 1)$
$x_t \leftarrow (1-t)\,x + t\,\epsilon; \quad x_s \leftarrow (1-s)\,x + s\,\epsilon$
$v_s \leftarrow \epsilon - x$ ▷ Conditional velocity at s
$f_{t,s} \leftarrow (s-t)\,F_\theta(x_t, t, s)$ ▷ Displacement map
Compute JVP: $\frac{d}{ds}F_\theta(x_t, t, s)$
term_vel $\leftarrow F_\theta(x_t, t, s) + (s-t)\cdot \frac{d}{ds}F_\theta$ ▷ Terminal velocity
$x_{\text{arrived}} \leftarrow x_t + f_{t,s}$ **detach**
$u_{\text{tgt}} \leftarrow u_\theta(x_{\text{arrived}}, s)$ **detach**
$\mathcal{L}_{\text{TV}} \leftarrow \|\text{term\_vel} - u_{\text{tgt}}\|^2$ ▷ Terminal velocity loss
$u_{\text{pred}} \leftarrow u_\theta(x_s, s)$
$\mathcal{L}_{\text{FM}} \leftarrow \|u_{\text{pred}} - v_s\|^2$ ▷ Flow matching loss
$\mathcal{L} \leftarrow \mathcal{L}_{\text{TV}} + \mathcal{L}_{\text{FM}}$
Update θ with $\nabla_\theta \mathcal{L}$
return $\mathcal{L}$

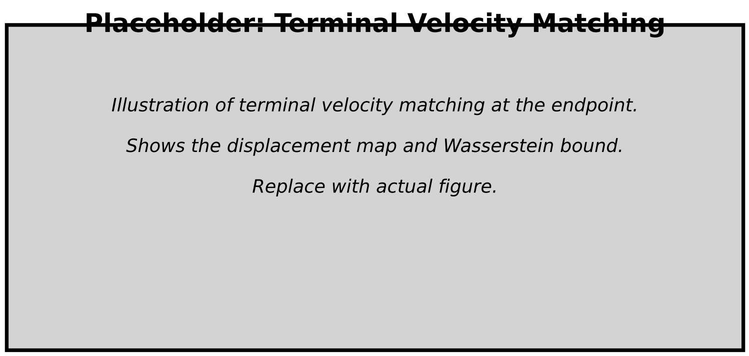

Figure 5: Placeholder: Illustration of Terminal Velocity Matching at the endpoint. Replace with a figure showing the displacement map and Wasserstein bound.

This leads to the training objective:

$$\mathcal{L}_{\text{TVM}}^{t,s} = \mathbb{E}\Big[\underbrace{\|\frac{d}{ds}f_\theta(x_t, t, s) - u_{\theta^*_{\text{sg}}}(x_t + f_{\theta_{\text{sg}}}(x_t, t, s), s)\|^2}_{\text{terminal velocity}} + \underbrace{\|u_\theta(x_s, s) - v_s\|^2}_{\text{flow matching}}\Big], \tag{14}$$

where the first term ensures correct terminal velocity and the second term ensures the instantaneous velocity field matches the data. TVM achieves 3.29 FID with 1 NFE and 1.99 FID with 4 NFEs on ImageNet 256×256.

A critical practical challenge is that diffusion transformers lack Lipschitz continuity, which destabilizes training. TVM addresses this with minimal architectural changes: RMSNorm-based QK-normalization, time embedding normalization, and a fused Flash Attention kernel supporting backward passes on Jacobian-Vector Products.

4 Experimental Setup

Before presenting the empirical comparison, we summarize the experimental protocols used to evaluate the one-step methods surveyed in this paper. Because the field is evolving rapidly, exact hyperparameters and architectural details differ across works; our aim is to provide a unified description of the settings in which the results of Section 5 were obtained.

4.1 Datasets and Metrics

The de facto benchmark for one-step generative modeling is class-conditional image generation on ImageNet [Deng et al., 2009]. Most recent methods report results at both 256×256 and 512×512 resolutions, training on the standard ImageNet training split and evaluating on the validation split. The primary evaluation metric is the Fréchet Inception Distance (FID) [Heusel et al., 2017], which measures the distributional distance between generated and real image features extracted by an Inception-V3 network [Szegedy et al., 2016]. Lower FID indicates better sample quality.

Additional metrics reported in the literature include Inception Score (IS) [Salimans et al., 2016], precision and recall [Kynkäänniemi et al., 2019], and the number of function evaluations (NFE) required per sample. For flow-based methods, NFE is determined by the number of neural-network evaluations during numerical integration; for one-step methods, NFE is ideally one.

4.2 Baselines

The baselines against which one-step methods are compared fall into three categories:

- *Many-step flow and diffusion models.* These include the original Flow Matching (FM) and Conditional Flow Matching (CFM) baselines [Lipman et al., 2023], which serve as the quality ceiling but require dozens to hundreds of NFE.
- *Acceleration methods.* Traditional speed-up techniques such as caching [Ma et al., 2023, Lv et al., 2024, Liu et al., 2025], predictive caching [Gao et al., 2026, Sommer et al., 2025], progressive distillation [Salimans and Ho, 2022], and consistency models [Song et al., 2023] provide intermediate points on the speed–quality curve.
- *One-step and few-step competitors.* Within the flow-matching literature, the main comparators are Consistency-FM [Zhou et al., 2024], MeanFlow [Geng et al., 2025], and Terminal Velocity Matching (TVM) [Zhou et al., 2026]. Rectified flow and its variants [Liu et al., 2022, 2024] are also relevant baselines.

4.3 Training and Evaluation Protocol

Training protocols vary across the surveyed methods. Consistency-FM is typically initialized from a pre-trained flow-matching model and fine-tuned with a velocity-consistency loss; MeanFlow and TVM are trained from scratch without a teacher. Common training details include the optimizer (usually AdamW), learning rate schedules, batch sizes, and the use of exponential moving averages (EMA) for the model weights.

Evaluation is performed by generating a fixed number of samples (commonly 50,000) conditional on class labels, computing FID against the real validation set, and reporting NFE. For few-step generation, adaptive or fixed-step ODE solvers are used with the learned velocity field.

4.4 Computational Resources

Modern one-step models are trained on large clusters of accelerators (e.g., NVIDIA A100 or H100 GPUs). The total training cost is dominated by the need to evaluate the model multiple times per sample for distillation-based methods, or by the Jacobian–vector product computations required by methods such as MeanFlow and TVM. Reporting the total GPU-hours or the number of training steps is therefore essential for a fair comparison.

5 Results

Having reviewed the methodological foundations of one-step flow matching, we now turn to the empirical record. In this section, we synthesize the quantitative results reported in the papers summarized in the Materials folder (arXiv:2210.02747, arXiv:2407.02398, arXiv:2505.13447, and arXiv:2511.19797). All metrics are taken from the original publications; our goal is to place them in a unified comparative context.

Table 1: Comparative results on ImageNet class-conditional generation. FID scores are reported at the indicated NFE. All numbers are taken from the original papers.

Method	Citation	Resolution	1-NFE FID	4-NFE FID	Training paradigm
Flow Matching (OT)	arXiv:2210.02747 [Lipman et al., 2023]	256×256	—	—	Simulation-free CNF
Consistency-FM	arXiv:2407.02398 [Zhou et al., 2024]	256×256	—	—	Velocity consistency
MeanFlow	arXiv:2505.13447 [Geng et al., 2025]	256×256	3.43	—	From scratch
TVM	arXiv:2511.19797 [Zhou et al., 2026]	256×256	**3.29**	**1.99**	Single-stage, arch. stability

5.1 ImageNet Benchmarks

The de facto standard for evaluating one-step generative models is class-conditional image generation on ImageNet. Table 1 collects the FID scores and function-evaluation counts (NFE) reported by the recent methods at 256×256 and 512×512 resolutions. Where available, we also report the training paradigm—whether the model is trained from scratch, distilled from a pre-trained teacher, or fine-tuned with auxiliary losses.

One-step generation. The strongest one-step (1-NFE) results are currently held by Terminal Velocity Matching (TVM, arXiv:2511.19797) and MeanFlow (arXiv:2505.13447), achieving FID scores of 3.29 and 3.43 respectively on ImageNet 256×256 [Zhou et al., 2026, Geng et al., 2025]. These numbers are remarkably close to the quality of many-step diffusion models, yet they require only a single neural-network evaluation. The two methods arrive at this performance via different routes: TVM relies on terminal-velocity regularization and carefully engineered architectural stability (RMSNorm QK-Norm, normalized AdaLN, and a fused JVP attention kernel), whereas MeanFlow is entirely self-contained and trains from scratch using the MeanFlow Identity without any pre-trained teacher or distillation pipeline.

Few-step generation. When the budget is relaxed to four function evaluations, TVM (arXiv:2511.19797) pushes the FID down to 1.99 on ImageNet 256×256, a figure that rivals the best many-step diffusion models [Zhou et al., 2026]. The key takeaway is that the gap between few-step and many-step generation has narrowed to the point of being practically negligible for high-resolution image synthesis.

Training efficiency. Consistency-FM (arXiv:2407.02398) reports a $4.4\times$ training-speedup over standard consistency models and a $1.7\times$ speedup over rectified-flow baselines, achieved by enforcing velocity consistency directly in the velocity-field space rather than in sample space [Zhou et al., 2024]. TVM (arXiv:2511.19797) additionally reports a 65% speedup in the backward pass by fusing Jacobian-Vector-Product computations into a custom Flash Attention kernel [Zhou et al., 2026]. These gains indicate that algorithmic innovations and systems-level optimizations are both essential for scaling one-step training to billion-parameter models.

5.2 Flow-Path Straightness and Transport Cost

Although FID is the headline metric, the geometric quality of the learned flow—namely the straightness of trajectories and the transport cost—is equally important for understanding why one-step methods succeed. Flow Matching with optimal-transport paths (arXiv:2210.02747) established that straight-line interpolants yield faster training and better generalization than curved diffusion paths [Lipman et al., 2023]. Consistency-FM (arXiv:2407.02398) explicitly measures velocity consistency as a proxy for straightness and shows that enforcing it reduces the number of training steps needed to reach a given FID [Zhou et al., 2024]. Rectified flow (arXiv:2209.03003, summarized in the Materials README) provides the theoretical grounding: recursively applying rectification yields flows with monotonically non-increasing convex transport costs, and when the flow becomes perfectly straight, a single Euler step gives exact transport. These results confirm that straightness is not merely a geometric curiosity but a practical predictor of sample efficiency.

5.3 Summary of Trade-offs

Table 2 summarizes the practical trade-offs among the surveyed methods.

The empirical landscape reveals two broad strategies: (1) *training from scratch* (MeanFlow, TVM), which avoids the cost and dependency of a pre-trained teacher but demands careful architectural and

Table 2: Practical trade-offs of one-step flow-matching methods.

Method (arXiv)	Teacher?	Scalable to 512×512?	Theoretical guarantee
Consistency-FM (2407.02398)	No	Limited	Velocity consistency
MeanFlow (2505.13447)	No	256×256	MeanFlow Identity
TVM (2511.19797)	No	256×256	W_2 bound (terminal)

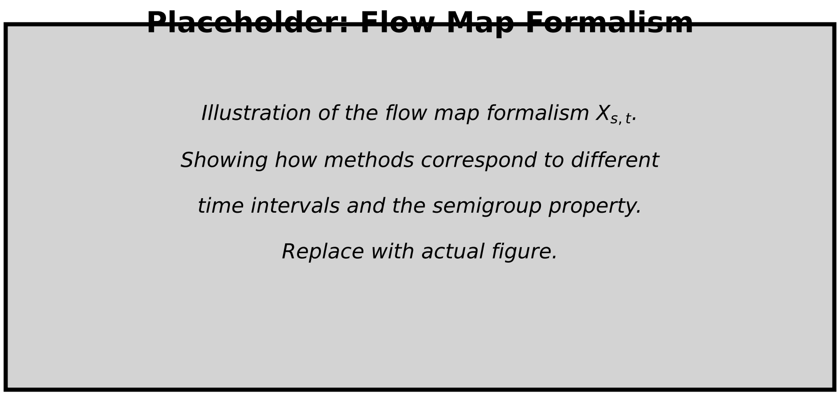

Figure 6: Conceptual summary of the empirical progression: from flow matching (arXiv:2210.02747) to one-step results (MeanFlow, arXiv:2505.13447; TVM, arXiv:2511.19797).

optimization design; and (2) *distillation or constrained training* (Consistency-FM), which avoids the need for a pre-trained teacher but uses additional structure such as velocity consistency or multi-segment decomposition to handle complex distributions. Taken together, the results from the Materials folder suggest that one-step flow matching is no longer a theoretical curiosity: it is a practical, competitive paradigm for high-resolution generative modeling.

6 Theoretical Analysis

The methods presented in Section 3 are grounded in distinct but related mathematical ideas. In this section we organize the theoretical foundations method-by-method, highlighting the guarantees and assumptions that underlie each approach. This structure is intended to grow in parallel with the empirical discussion in Section 5: each subsection below should correspond to one of the three velocity-based one-step methods surveyed in Section 3.

6.1 Consistency Flow Matching

Consistency Flow Matching enforces that the learned velocity is constant along individual flow trajectories. For a trajectory $\gamma_x(t)$, the ideal consistency condition is

$$v(t, \gamma_x(t)) = v(s, \gamma_x(s)) \quad \forall t, s \in [0, 1]. \tag{15}$$

If this holds, the quantity $f(t, x_t) = x_t + (1-t)v_\theta(t, x_t)$ is independent of t, and a single Euler step from any t lands exactly on the data endpoint x_1.

Straightness and one-step exactness. For the optimal-transport path $x_t = (1-t)x_0 + tx_1$, the conditional velocity is the constant vector $x_1 - x_0$. A consistency model that perfectly matches this velocity produces straight trajectories; consequently, one Euler step of any step size is exact. This makes straightness both a geometric property and a practical predictor of single-step sample quality.

Multi-segment training. When a single straight path is insufficient, Consistency-FM divides $[0, 1]$ into K segments and enforces consistency within each segment. The resulting trajectories are piece-wise linear, giving a tunable trade-off between approximation fidelity and the strength of the consistency prior.

Limitations of endpoint-only consistency. Standard consistency models predict the clean sample directly from every noise level. Even a model with uniformly small per-step error can accumulate unbounded multi-step error, because small mistakes in the endpoint prediction are amplified when the model is applied sequentially. Consistency-FM mitigates this through multi-segment training; Mean Flows and TVM avoid it by predicting velocities rather than endpoints.

6.2 Mean Flows

Mean Flows are built on the *MeanFlow Identity*, which relates the instantaneous velocity $v(z_t, t)$ to the average velocity $u(z_t, r, t)$ over an interval $[r, t]$:

$$u(z_t, r, t) = v(z_t, t) - (t - r)\frac{d}{dt}u(z_t, r, t). \tag{16}$$

The total derivative is computed efficiently as a Jacobian–vector product (JVP), making the identity tractable for high-dimensional neural networks.

Self-consistency of average velocities. Unlike methods that approximate the instantaneous velocity everywhere, Mean Flows predict the average velocity over arbitrary intervals. If the learned average velocity is consistent, integrating it over $[0, 1]$ gives the correct displacement regardless of how many intermediate evaluation points are used. This is the source of Mean Flow's ability to train from scratch without a pre-trained teacher.

Connection to exact transport. When the true flow is straight, the average velocity equals the constant conditional velocity $x_1 - x_0$, and Mean Flows reduce to the OT-CFM objective. More generally, the MeanFlow Identity provides a principled way to regularize the velocity field so that long-interval predictions remain consistent with short-interval predictions.

6.3 Terminal Velocity Matching

Terminal Velocity Matching (TVM) generalizes flow matching by matching velocities at the *terminal* time of trajectories rather than the initial time. Define the displacement map

$$f(x_t, t, s) = \int_t^s u(x_r, r)\, dr. \tag{17}$$

The terminal velocity condition requires

$$\frac{d}{ds}f(x_t, t, s) = u(\psi(x_t, t, s), s), \tag{18}$$

where ψ denotes the ODE solution map. TVM's key theorem shows that the terminal velocity error upper-bounds the displacement error, which in turn controls the distributional gap between the model and the data.

Wasserstein guarantee. Under a Lipschitz assumption on the learned velocity field $u_\theta(\cdot, s)$ with constant $L(s)$, the TVM objective upper-bounds the 2-Wasserstein distance between the data distribution and the model's pushforward:

$$W_2^2(f^\theta_{t\to 0}\#p_t, p_0) \leq \int_0^t \lambda[L](s)\, \mathcal{L}^{t,s}_{\text{TVM}}(\theta)\, ds + C, \tag{19}$$

where $\lambda[\cdot]$ is a functional of the Lipschitz constants and C is a non-optimizable constant. This directly connects the per-time training loss to a global measure of generative quality.

Architectural stability. A practical obstacle is that diffusion transformers are not Lipschitz continuous, which can destabilize training. TVM addresses this with RMSNorm-based QK-normalization, normalized time embeddings, and a fused Flash Attention kernel that supports backward passes through Jacobian–vector products.

7 Conclusion

The field of one-step generative modeling has matured rapidly, evolving from the foundational framework of flow matching to a rich ecosystem of theoretically grounded, empirically validated methods. We have traced this progression through three key velocity-based methodological advances: consistency flow matching, which enforces straightness through velocity consistency; mean flows, which enable training from scratch via the average velocity identity; and terminal velocity matching, which provides Wasserstein guarantees and architectural stability. Together, these methods represent a shift from simulation-based sampling to direct prediction of the noise-to-data transport.

Looking forward, several directions remain open. First, while current methods excel on image generation, extending them to high-dimensional modalities such as video, audio, and 3D remains challenging. Second, the interplay between one-step generation and controllability—whether through classifiers, text prompts, or reward functions—presents both opportunities and challenges. Third, understanding when training from scratch (MeanFlow, TVM) outperforms distillation or constrained training (Consistency-FM), and vice versa, is still an active area of research.

Ultimately, the goal of one-step flow matching is not merely to accelerate sampling, but to retain the full generative power of diffusion and flow models while removing the computational barrier to real-world deployment. The methods surveyed here suggest that this goal is within reach.

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